

Rui (Betty) Zhou

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EDUCATION

Brown University *M.S. in Computer Science*, Overall GPA: 4.0/4.0 09/2024-05/2026(expected)

Core Courses: Computer Vision, Deep Learning, Software Engineering, Intelligent Robotics

Harvard University 09/2024-12/2024

Visiting coursework: Innovation in Science and Engineering

University of California, Davis 09/2020 - 04/2024

B. S. in Computer Science, Minor in Managerial Economics, Overall GPA: 3.83/4.0

Core Courses: Algorithms, Data Structures, OOD, Operating Systems, Networks, Web Dev, ML, Computer Graphics

EXPERIENCE

Brown University, Brown Visual Computing Group Providence, RI, USA

Research Assistant 09/2024 – present

- Co-authored *WACV 2026* accepted paper, [GenHSI: Controllable Generation of Human-Scene Interaction Videos](#), proposing a training-free 3D-aware keyframe prompting pipeline for long, controllable HSI video generation.

- Managed datasets for video generation experiments; implemented automated data validation and preprocessing scripts in Python.

- Leading body2GarmentCode, a program synthesis project that infers parametric GarmentCode programs from designer-created sewing patterns and body measurement variants, enabling generalized garment grading across body shapes.

Style3D Hangzhou, Zhejiang, China

Software Engineer Intern (AI/ML) 04/2024 – 07/2024

- Built production ML pipelines for Fabric-TryOn feature, processing and transforming image datasets through Python-based ETL workflows with Stable Diffusion and PyTorch. The upgraded product achieved 850K+ units sold and adoption by 60+ enterprise clients.
- Developed a texture inpainting/outpainting module for seamless fabric rendering.

- Conducted data analysis across multimodal embedding models (InternVL, IXC-2.5, Florence-2) to improve Chinese-language text-to-image alignment. Presented research on model performance to the team, accelerating adoption of optimized pipelines.

UC Davis, Data Visualization Lab Davis, CA, USA

Research Assistant 01/2023 – 03/2024

- Architected and deployed an end-to-end RAG data pipeline for the [GreenMine Project](#), processing 600+ pages of unstructured legal/interview data into structured, queryable formats using GPT-4 and Qdrant vector database.

- Designed scalable data ingestion and embedding workflows with multi-threading, improving query response time by 25% and enabling real-time semantic search for stakeholders.

PROJECTS

PDFeditor (Full-Stack Document Translation App) 09/2025 – 11/2025

- Built an end-to-end full-stack app with a React & TypeScript editor panel for interactive translated content—users can move, resize, and edit overlaid text directly on the original PDF layout; authenticated via Firebase with NoSQL document storage.

- Designed a backend translation pipeline chaining MinerU for structured PDF parsing and GPT-4o for context-aware translation, with RESTful APIs handling document ingestion, glossary injection, and rendered output delivery.

DINOLIP (Vision-Language Compositional Reasoning) 10/2025 – 12/2025

- Trained two CLIP-style dual-encoder architectures using DINOv3 as a frozen visual backbone on CC3M and CC12M.

- Built a synthetic evaluation pipeline generating 1,000+ structured image triplets with attribute and relation swaps; defined a binding sensitivity metric to isolate encoder capacity from cross-modal alignment effects.

- Fine-tuned DinoLIP model; identified overfitting and encoder inductive bias as primary bottlenecks to compositional generalization, informing future architecture directions.

TECHNICAL SKILLS

Programming Languages: Python, Typescript, C/C++, Java, R; **Languages:** English (Full Professional), Chinese (Native)

Frameworks/Tools: Linux, NoSQL, PyTorch, TensorFlow, OpenCV, PIL, NumPy, Pandas, Git, GDB, Vue JS, Qdrant, MeshLab;